

torch.addr#

<https://pytorch.org/docs/stable/generated/torch.addr.html#torch.addr>

Description:

Performs the outer-product of vectors `vec1` and `vec2` and adds it to the matrix input. Optional values `beta` and `alpha` are scaling factors on the outer product between `vec1` and `vec2` and the added matrix input respectively. If `beta` is 0, then the content of input will be ignored, and `nan` and `inf` in it will not be propagated. If `vec1` is a vector of size `n` and `vec2` is a vector of size `m`, then input must be broadcastable with a matrix of size $(n \times m)(n \times m)(n \times m)$ and out will be a matrix of size $(n \times m)(n \times m)(n \times m)$. input (Tensor) – matrix to be added

Function Signature

```
torch.addr(input, vec1, vec2, *, beta=1, alpha=1, out=None)
→ Tensor#
```

Parameters

Keyword Arguments

Parameters

Keyword

beta (Number, optional) – multiplier for input (β) alpha (Number, optional) – multiplier for $\text{vec1} \otimes \text{vec2}$ (α) out (Tensor, optional) – the output tensor.

Code Examples

```
>>> vec1 = torch.arange(1., 4.)
>>> vec2 = torch.arange(1., 3.)
>>> M = torch.zeros(3, 2)
>>> torch.addr(M, vec1, vec2)
tensor([[ 1.,  2.],
        [ 2.,  4.],
        [ 3.,  6.]])
```

Detailed Documentation

Documentation

Performs the outer-product of vectors `vec1` and `vec2` and adds it to the matrix input.

Optional values `beta` and `alpha` are scaling factors on the outer product between `vec1` and `vec2` and the added matrix input respectively.

If `beta` is 0, then the content of input will be ignored, and `nan` and `inf` in it will not be propagated.

If `vec1` is a vector of size `n` and `vec2` is a vector of size `m`, then input must be broadcastable with a matrix of size $(n \times m)(n \times m)(n \times m)$ and out will be a matrix of size $(n \times m)(n \times m)(n \times m)$.

`input` (Tensor) – matrix to be added

`vec1` (Tensor) – the first vector of the outer product

`vec2` (Tensor) – the second vector of the outer product

`beta` (Number, optional) – multiplier for input (β)

`alpha` (Number, optional) – multiplier for $\text{vec1} \otimes \text{vec2}$ (α)

`out` (Tensor, optional) – the output tensor.

